

TASK 4: DASHBOARD DEVELOPMENT AND VISUALIZATION

1.Introduction

After completing data cleaning and exploratory data analysis, dashboards were developed using Microsoft Excel and Power BI to transform raw data into meaningful business insights. These dashboards help monitor sales performance, customer behavior, product revenue, referral sources, and payment methods through interactive visualizations.

2.Objective

- Develop interactive dashboards for business performance monitoring.
- Track Key Performance Indicators (KPIs).
- Visualize sales and revenue trends.
- Identify top-performing products and referral sources.
- Support data-driven decision making.

3. Excel Dashboard Development

Tools Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Slicers
- Conditional Formatting

Dashboard Components

KPI Cards

- Total Revenue
- Total Orders
- Average Order Value
- Total Quantity Sold
- Total Customers

Visualizations Created

- Revenue by Product
- Revenue by Referral Source
- Monthly Revenue Trend
- Orders by Payment Method
- Product-wise Revenue Analysis

Interactive Features

- Year Slicer
- Month Slicer
- Product Filter
- Dynamic Pivot Charts

Excel Dashboard Development Process

1. Imported cleaned dataset into Excel.
2. Created Pivot Tables for analysis.
3. Generated Pivot Charts for visual representation.
4. Added slicers for interactive filtering.
5. Designed KPI cards using formulas and formatting.

6. Organized all visuals into a single dashboard layout.

Results Obtained

The Excel dashboard provided a clear overview of sales performance and enabled quick analysis of business metrics through filters and charts.

4.Power BI Dashboard Development

Tools Used

- Power BI Desktop
- DAX
- Data Modeling
- Power Query

Dashboard Components

KPI Cards

- Total Revenue
- Total Orders
- Average Order Value
- Total Quantity
- Total Customers

Visualizations

Revenue by Product

Treemap showing revenue contribution of each product category.

Revenue by Referral Source

Donut Chart displaying revenue generated from different referral sources.

Monthly Revenue Trend

Line Chart showing monthly revenue fluctuations.

Year Revenue Trend

Line Chart displaying yearly revenue performance.

Orders by Payment Method

Bar Chart representing customer payment preferences.

Filters

- Year Filter
- Month Filter
- Date Range Filter

DAX Measures Used

Best Product:

```
Best Product =  
TOPN(  
1,  
VALUES(Table1[Product]),  
CALCULATE(SUM(Table1[TotalPrice])))  
)
```

Best Referral:

```
Best Referral =  
TOPN(  
1,  
VALUES(Table1[ReferralSource]),  
CALCULATE(SUM(Table1[TotalPrice])))
```

)

Best Month:

Best Month =

"June"

Results

- Total Orders: 1200
- Total Revenue: ₹12,64,761.96
- Average Order Value: ₹1053.97
- Best Product: Chair
- Product Revenue: ₹195,620.11
- Best Referral Source: Instagram
- Referral Revenue: ₹275,285.45
- Best Year: 2023
- Best Year Revenue: ₹552,643.24
- Best Month: June 2024
- Best Month Revenue: ₹68,068.54

5.Key Insights

- Chair generated the highest revenue among all products.
- Instagram delivered the maximum revenue and customer acquisition.
- Year 2023 recorded the highest overall revenue.
- June 2024 was the best-performing month.
- Online payment method was the most preferred payment option.
- Revenue trends highlighted seasonal fluctuations in sales performance.

6.Conclusion

The Excel and Power BI dashboards successfully transformed raw e-commerce transaction data into meaningful business insights. Interactive visualizations and KPI tracking enabled efficient performance monitoring and data-driven decision-making. The dashboards provide management with a comprehensive view of sales trends, customer behavior, and business performance.

7. Dashboard Screenshots

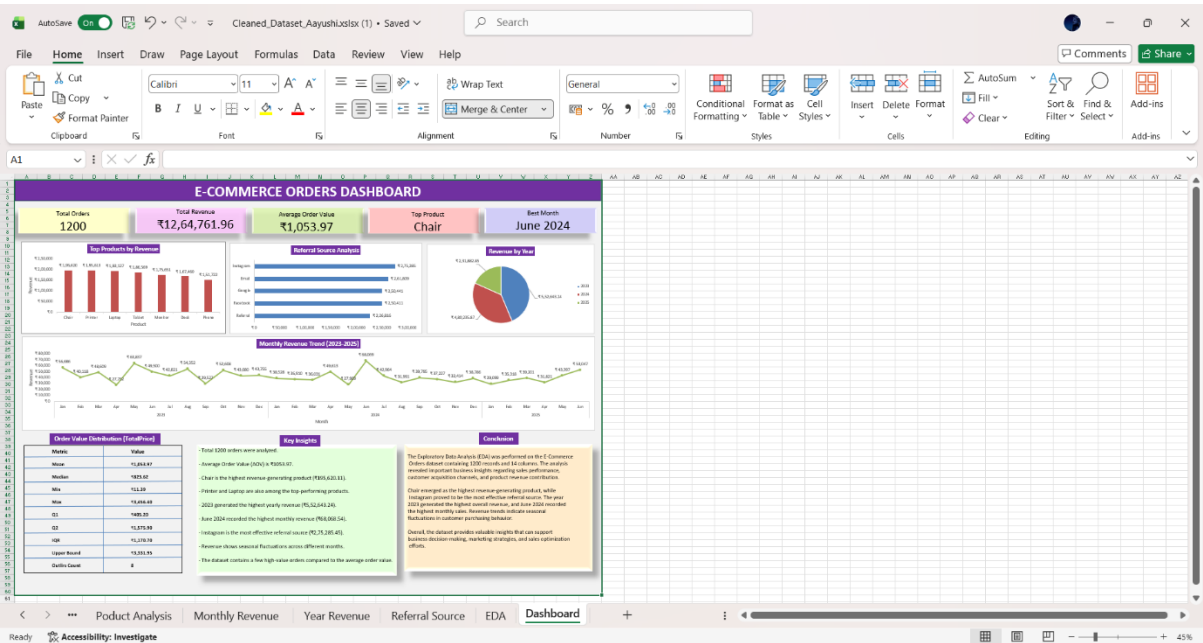


Figure 1: Excel Dashboard

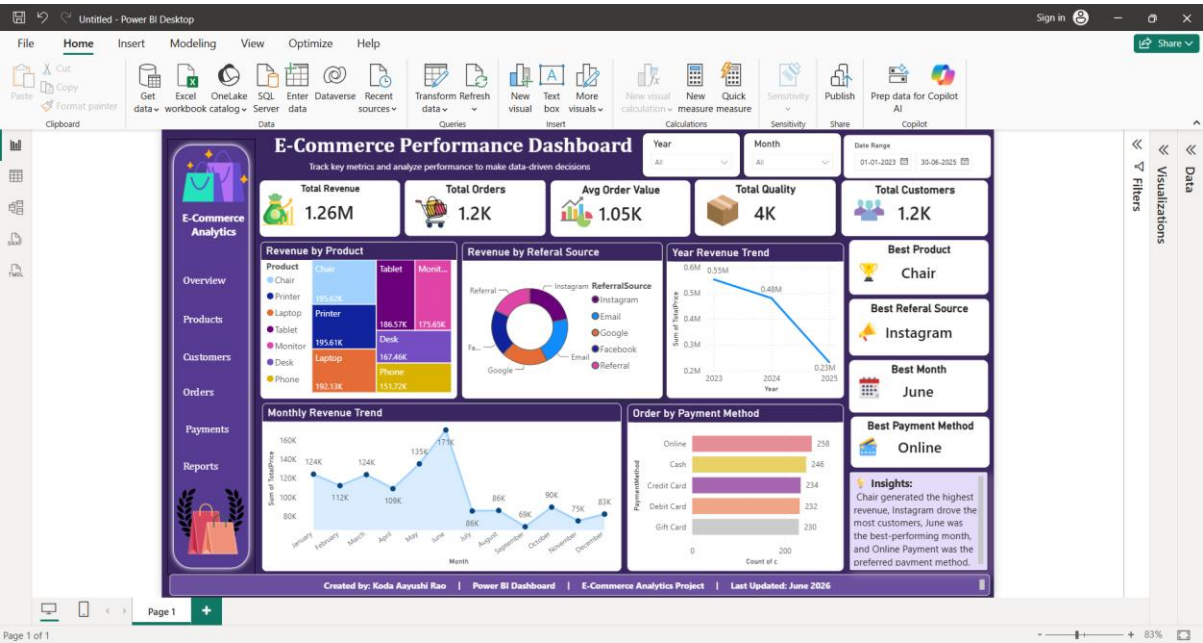


Figure 2: Power BI Dashboard

